

Clément Sebastiao

* 23 November 2000 • ✉ clse@itu.dk • 🌐 csebastiao.github.io
in clément-sebastiao-530455315 • 🌐 csebastiao
🆔 0009-0009-3084-0071 • 📄 9DteMXgAAAAJ

Experience

IT University of Copenhagen

Copenhagen, Denmark

PhD Student

Apr 2024–Present

Optimizing urban sustainable mobility networks, part of the Just Streets EU Horizon Program (Grant agreement ID: 101104240). Supervised by Michael Szell and Luca Maria Aiello.

MIT Senseable City Lab

Cambridge, Massachusetts, USA

Visiting Student

Sep 2025–Feb 2026

Gender-Age Mobility Disparities in Tokyo. Supervised by Paolo Santi.

IT University of Copenhagen

Copenhagen, Denmark

Research Assistant

Jan 2025–April 2025

Course Manager and Lecturer of the Master course "Geospatial Data Science".

IT University of Copenhagen

Copenhagen, Denmark

Research Assistant

Jan 2024–Mar 2024

Université de Bourgogne-Franche-Comté

Besançon, France

Intern

Feb 2023–Jul 2023

Mode of Transport and Historical Road Network Growth Relationship, supervised by Claire Lagesse.

ISI

Turin, Italy

Intern

Oct 2022–Jan 2023

Nature and Bicycle Network Relationship, supervised by Rossano Schifanella.

IT University of Copenhagen

Copenhagen, Denmark

Intern

Feb 2022–Jul 2022

Prioritizing Links in Bicycle Network Plans With Subtractive Network Optimization, supervised by Michael Szell.

LIPHY

Grenoble, France

Intern

May 2021–Jul 2021

Collective Motion of Schooling Fish in a Complex Environment, supervised by Aurélie Dupont and Philippe Peyla.

Ecole Normale Supérieure de Lyon

Lyon, France

Intern

Jun 2020–Jul 2020

Deep Learning for Time Series Predictions for FECG Monitoring, supervised by Pierre Borgnat and Patrice Abry.

Université d'Aix-Marseille

Marseille, France

Intern

Jun 2019–Jul 2019

Study of Optically Excited Magnetic States of ZnO by EPR, supervised by Adrien Savoyant.

Education

Ecole Normale Supérieure de Lyon <i>Master degree, ENS Lyon Diploma,</i>	Lyon, France <i>2019–2023</i>
Ecole Normale Supérieure de Lyon <i>Master degree in Complex Systems, Physics,</i>	Lyon, France <i>2020–2022</i>
Claude Bernard Université Lyon 1 <i>Bachelor degree in Physics,</i>	Lyon, France <i>2017–2020</i>

Languages

French: Native speaker

English: Proficient, A1/A2 *Cambridge Advanced 199/210*

Skills

Research and Productivity Software: QGIS, LaTeX, Visual Studio Code, Zotero, Git.

Python: Network Science (networkx, igraph), Data Science (pandas, numpy, scipy), Geospatial analysis (geopandas, OSMnx, shapely, momepy), data visualization (matplotlib, seaborn, folium).

Other programming languages (decreasing order of proficiency): R, PostgreSQL, Matlab

Research Interests

Primary: Urban Data Science, Sustainable and Just mobility, Complex Systems

Secondary: Applied Network Science, Data visualization

Peer-reviewed Articles

[1] Adrien Savoyant, Clément Sebastiao, and Olivier Margeat. Long-Standing Optically Excited Magnetic States in ZnO Nanoparticles. *physica status solidi (RRL) – Rapid Research Letters*, 14(7):2000176, 2020.

[2] Clément Sebastiao and Michael Szell. Right-wing Votes Relate to Delays in Bicycle Network Development. *Findings*, July 2026.

[3] Clément Sebastiao, Anastassia Vybornova, Ane Rahbek Vierø, Luca Maria Aiello, and Michael Szell. The trade-off between directness and coverage in transport network growth. *Applied Network Science*, 11(1):34, March 2026.

Teaching

Courses.....

IT University of Copenhagen <i>Geospatial Data Science, 7.5 ECTS</i> Course Manager and Lecturer (100%)	Copenhagen, Denmark <i>Feb 2025 – Jul 2025</i>
--	--

Single classes.....

IT University of Copenhagen <i>IT and Green Transitions, 7.5 ECTS</i> Urban Data Science and Sustainable Mobility	Copenhagen, Denmark <i>Sep 2026</i>
IT University of Copenhagen <i>IT and Green Transitions, 7.5 ECTS</i> Urban Data Science and Sustainable Mobility	Copenhagen, Denmark <i>Oct 2024</i>
IT University of Copenhagen <i>Geospatial Data Science, 7.5 ECTS</i> Bicycle networks	Copenhagen, Denmark <i>Jun 2022</i>

Conferences and talks

D3A 4.0 <i>Talk</i> Ordering the growth of urban bicycle networks	Nyborg, Denmark <i>Oct 2026</i>
Network Science Institute <i>Talk</i> Leveraging Network Science to Promote a More Just, Sustainable, and Active Mobility	Boston, Massachusetts, USA <i>Jan 2026</i>
Conference on Complex Systems 2025 <i>Talk</i> The Trade-Off between Directness and Coverage in Planar Network Growth	Siena, Italy <i>Sep 2025</i>
Just Streets <i>Talk</i> Street Space Reallocation: Superblockify	Online <i>May 2025</i>
D3A 2.0 <i>Poster</i> Prioritizing links in bicycle network plans	Nyborg, Denmark <i>Oct 2024</i>
Conference on Complex Systems 2024 <i>Talk</i> Prioritizing links in bicycle network plans with greedy network optimization	Exeter, UK <i>Sep 2024</i>
UrbanSys2024 <i>Poster</i> Translating active mobility goals into network metrics	Exeter, UK <i>Sep 2024</i>
MIT Senseable City Lab <i>Talk</i> Pecha Kucha: Bicycle Plans and Superblockify	Online <i>Aug 2024</i>
Nordic Geographers Meeting 2024 <i>Talk</i> Prioritizing links in bicycle network plans	Copenhagen, Denmark <i>Jun 2024</i>
PRIMS <i>Talk</i> Morphogénèse urbaine	Arc-et-Senans, France <i>Nov 2023</i>
Rencontres Francophones Transport Mobilité 2023 <i>Talk</i> Inventaire et analyse de motifs viaires pour une compréhension du développement urbain sur le temps long	Dijon, France <i>Jun 2023</i>

LIPHY <i>Lightning Talk</i> Effets collectifs en environnement complexe chez les poissons	Grenoble, France <i>Jun 2022</i>
Conference on Complex Systems 2021 <i>Attending</i>	Lyon, France <i>Oct 2021</i>

Summer schools and advanced courses

Umeå University <i>Icelab Camp</i>	Umeå, Sweden <i>Sep - Oct 2026</i>
Santa Fe Institute <i>Complex Systems Summer School</i>	Santa Fe, New Mexico, USA <i>Jun 2025 – Jul 2025</i>
Royal Danish Academy <i>Sustainable Futures IV (PhD Course)</i>	Copenhagen, Denmark <i>Oct 2024 – Jan 2025</i>
Aalborg University <i>The Anatomy of Mobility Injustice (PhD Course)</i>	Aalborg, Denmark <i>Sep 2024</i>
Fondazione Alessandro Volta <i>Advances in Complex Systems Summer School</i>	Como, Italy <i>Jul 2024</i>

Academic services

Reviews	
Scientific Data: 2024	
Nature Cities: 2026	
Organization and committee	
D3A 4.0 <i>Workshop Co-organizer</i>	Nyborg, Denmark <i>Oct 2026</i>
CS2Nordics 2026 <i>Program Committee</i>	Copenhagen, Denmark <i>Sep 2026</i>
Eurodoc Conference 2025 <i>Volunteer</i>	Copenhagen, Denmark <i>Aug 2025</i>
ITU PhD Club <i>Treasurer, Committee Member</i>	Copenhagen, Denmark <i>Nov 2024–Nov 2026</i>

Academic Awards

UrbanSys2024 <i>Best poster</i>	Exeter, UK <i>Sep 2024</i>
---	--------------------------------------